

Sam Degnan

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Summary

Software Engineer focused on secure embedded systems, cryptography, cloud infrastructure, and developer tooling. Builds security-sensitive software across embedded C/C++, AWS infrastructure, CI/CD, and AI-assisted development workflows. Experienced in taking features from design through implementation, testing, release, and documentation.

Work Experience

Motorola Solutions | Software Engineer I

June 2023 – Present

- Engineered FIPS-compliant cryptographic software for first-responder networks and served as the team's Cyber Security Champion, implementing zero-trust protocols and secure key management.
- Evaluated AI-generated code for security risks and enforced output validation in a secure communications environment, preventing sensitive material from being exposed through AI tooling.
- Championed the integration of AI coding assistants in a secure communications environment, cutting month-long feature cycles down to one week, a 75% speedup.
- Developed an automated framework to accelerate test scaffolding and integrated SonarQube and Klocwork secure-coding checks into CI pipelines.
- Programmed and debugged embedded C/C++ on ARM Cortex-M and STM32 RTOS targets, using Lauterbach JTAG to squeeze performance out of tightly constrained hardware.
- Modernized team DevOps workflows by standing up AWS infrastructure (EC2, IAM, Secrets Manager) and containerizing builds with Docker, Jenkins, and Bitbucket CI.
- Owned end-to-end release management for packaged applications and drivers using InstallShield, Conan, and TestRail, while managing code-signing workflows.
- Authored technical documentation, unit tests, and interface specifications that reduced onboarding time and long-term maintenance burden.

Selected GitHub Projects

- **Zero-Knowledge Fitness Platform:** Built a PWA fitness tracker with a Node.js/Express and PostgreSQL backend, using AES-256-GCM client-side encryption, IndexedDB persistence, and optional ML-KEM/ML-DSA authenticated sync with a scoped Groth16 proof layer.
- **Secure Ephemeral Environment Orchestrator (SEEO):** Engineered a React/Rails 7 platform that provisions isolated ephemeral dev environments using Terraform, AWS, Docker, and OPA/Rego.
- **AI-Driven CI/CD Security Guardrail:** Developed a Python tool that leverages multiple LLMs to analyze pull requests, outputting SARIF findings and enforcing OPA/Rego policies via GitHub Actions.

Technical Skills

- **Languages:** C, C++, Python, Java, C#, Ruby, JavaScript, SQL, Bash
- **Security & Embedded:** FIPS 140, Key Management, Code Signing, OpenSSL, STM32, ARM Cortex-M, RTOS, JTAG, SPI/I2C
- **Cloud & DevOps:** AWS (EC2, AMIs, IAM, Secrets Manager), Docker, Docker Compose, Terraform, GitHub Actions, Jenkins, CI/CD, CMake, Make, Azure DevOps
- **Tools & Workflows:** React, Ruby on Rails, PostgreSQL, IndexedDB, Playwright, SARIF, OPA/Rego, AI-Assisted Development, LLM Tooling, Git, SonarQube, Klocwork, Linux/UNIX, VS Code

Education

Iowa State University of Science and Technology | **B.S. Software Engineering** | August 2019 - May 2023